Journal of Soil and Rhizosphere Biotechnology
https://www.acadlore.com/journals/JSRB

An Exploratory Comparison of Target-Gene Signals for Selected Bacterial and Fungal Taxa in Agricultural and Forest Soils in Panchkhal, Nepal

Aakriti Adhikari, Yanu Bam, Diwash Gautam, Siddhartha Kafle, Divya Panta, Amol Dahal*

Department of Biotechnology, Kathmandu University, 45210 Dhulikhel, Nepal

* Correspondence Amol Dahal (amol.dahal@ku.edu.np)

Received: 04-10-2026 **Revised:** 05-08-2026 **Accepted:** 05-15-2026

Citation: Adhikari, A., Bam, Y., Gautam, D., Kafle, S., Panta, D., & Dahal, A. (2026). An exploratory comparison of target-gene signals for selected bacterial and fungal taxa in agricultural and forest soils in Panchkhal, Nepal. *J. Soil Rhizosphere Biotechnol.*, *1*(1), 2–10. https://doi.org/10.56578/jsrb010102.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: The intensification of global agriculture has increased reliance on agrochemicals that may alter soil microbial communities. However, differences in selected soil microbial taxa between agricultural and forest sites remain poorly characterized in Nepal. This exploratory study used quantitative polymerase chain reaction (qPCR) to compare target-gene signals for selected bacterial and fungal taxa between one agrochemically managed agricultural field and one nearby forest site in Panchkhal, Nepal. The total bacterial signal was estimated to be 0.84-fold that of the forest soil, but the difference was not statistically significant ($p = 0.531$). Among the bacterial taxa, the signals for *Alphaproteobacteria* and *Acidobacteria* were lower in agricultural soil, with estimated fold differences of 0.57 ($p = 0.002$) and 0.26 ($p = 0.008$), respectively. In contrast, the signal for *Betaproteobacteria* was 1.97-fold higher in agricultural soil ($p = 0.001$), whereas the signal for *Actinobacteria* was 1.09-fold higher, with no statistically significant difference ($p = 0.768$). The total fungal signal was substantially lower in agricultural soil (estimated fold difference = 0.06, $p = 0.007$). The signal for *Ascomycota* was also lower (estimated fold difference = 0.40; reported $p = 0.0503$), while *Basidiomycota*-specific amplification was not detected in agricultural soil under the applied qPCR conditions. These findings provide preliminary, site-specific evidence of differences in qPCR-derived target-gene signals between the two sampled sites. Further research involving multiple independent sites, soil physicochemical characterization, and agrochemical residue measurements is needed to investigate the factors associated with these differences.

Keywords: Quantitative polymerase chain reaction; Agrochemicals; Soil health; Soil microbial taxa; Target-gene signals; Agricultural soils

1. Introduction

The intensification of global agriculture has led to a heavy reliance on agrochemicals, mainly synthetic fertilizers and pesticides, to maximize crop yields and protect plants from biological threats. The use of these chemicals has become a primary input to tackle pests and prevent crop failures, with consumption increasing at an annual rate of 10–20% in Nepal (Ghimire & GC, 2018). While these inputs contribute to national food security by reducing harvest losses, their indiscriminate and heavy application has brought about substantial negative impacts on human health, the ecosystem, and soil microbial communities (Paudel et al., 2024). In regions like Panchkhal, Kavre district of Nepal, farmers have reported acute health issues such as respiratory problems, skin rashes, and chest pain, while chronic exposure is associated with potential adverse health effects, including cancer, neurotoxicity, and birth defects (GC & Neupane, 2019; Sapkota et al., 2020). Environmentally, the overuse of synthetic fertilizers can lead to chemical leaching into groundwater and a fundamental disruption of the soil's biological and chemical properties (Srivastav, 2020).

Microbial diversity is the cornerstone of soil health, representing the "living skin" of the earth and one of the most complex habitats on the planet. These microorganisms are the primary drivers of biogeochemical cycles, decomposing organic matter to release vital nutrients like nitrogen, phosphorus, and carbon back into the biosphere

(Osinuga et al., 2023). Beyond nutrient cycling, the soil microbiome controls plant immune responses, suppresses pathogens, and increases host resistance to abiotic stresses such as drought. Furthermore, soil microorganisms contribute to soil carbon sequestration through the accumulation of microbial biomass and necromass (Sae-Tun et al., 2022). However, the heavy application of agrochemicals can lead to soil dysbiosis, where high levels of synthetic nitrogen and pesticides potentially inhibit beneficial nitrogen-fixing bacteria and fungi, thereby altering microbial abundance and community composition (Némethová et al., 2026).

Maintaining soil health through sustainable agriculture is essential for long-term productivity, as the prolonged overuse of agrochemicals can eventually reduce the efficiency of crop yields (Osinuga et al., 2023). While traditional soil mapping in Nepal, conducted by the National Soil Science Research Center (NSSRC) under Nepal Agricultural Research Council (NARC), has focused on physico-chemical parameters like pH and nutrient levels, there is an urgent need to prioritize soil biodiversity as a functional, living infrastructure. Understanding differences in soil microbial communities between agricultural and forest sites can provide preliminary evidence to inform future investigations of soil health and agricultural management. Despite this imperative, studies comparing quantitative polymerase chain reaction (qPCR)-derived signals for selected soil microbial taxa between agricultural and forest sites in Nepal remain limited. Although high-intensity farming areas like Panchkhal are known for heavy pesticide consumption (Paudel et al., 2024), molecular evidence describing soil microbial communities in these agroecosystems remains scarce. To our knowledge, no published study has compared target-gene signals for selected bacterial and fungal taxa between an agrochemically managed agricultural field and a forest site in Nepal.

Consequently, this exploratory study aimed to compare qPCR-derived signals for selected bacterial and fungal taxa between one agrochemically managed agricultural field and one nearby forest site in Panchkhal, Nepal. The study employed qPCR to measure target-specific amplification signals and estimate differences between the two soil samples using the Ct-difference approach ($2^{-\Delta Ct}$). The findings provide preliminary, site-specific data that may serve as a foundation for future investigations incorporating multiple independent sampling sites, comprehensive soil physicochemical characterization, pesticide residue analyses, and high-resolution microbial community profiling.

2. Methodology

2.1 Soil Sampling

Soil samples were collected at a depth of approximately 15 cm below the soil surface at a forest site (27°39'34.9"N, 85°37'05.5"E; elevation 886 m) and an agrochemically treated cauliflower farm (27°38'59.7"N, 85°37'04.8"E; elevation 862 m) covering approximately 0.9 acres of land in Panchkhal, Nepal. Soil sampling was conducted on 25 November 2025, one day after the application of a commercial insecticide, Cora Plus™ +++ (Emamectin Benzoate 2% suspension concentrate + Indoxacarb 8%), at a concentration of 0.5 mL/L of water, and two fungicides, Bavistin (carbendazim 50% wettable powder) and Indofil M-45 (mancozeb 75% wettable powder), each applied at a concentration of 2 g/L of water, to the agricultural farm.

Soil samples were collected from five randomly selected locations within each sampling site (forest and agricultural field). Each soil subsample was placed in a sterile zip-lock bag, appropriately labeled, transported to the laboratory in an ice box, and processed immediately upon arrival. For each composite sample, 30 g of soil from each of the five subsamples was weighed and thoroughly homogenized to produce a 150 g composite sample. This sampling procedure was repeated independently using a different set of five randomly selected locations, resulting in two separately collected composite samples (within-site sampling replicates) for each site.

2.2 DNA Extraction and Quantification

Total soil DNA was extracted from 0.2 ± 0.05 g of each composite soil sample using the PureLink™ Microbiome DNA Purification Kit (Invitrogen, Thermo Fisher Scientific India Pvt Ltd., Mumbai, India; Cat. No. A29790), following the manufacturer's instructions for DNA extraction. The extraction process included homogenization of soil with a bead-beating lysis buffer, removal of inhibitors using a cleanup buffer and centrifugation, binding of DNA to a spin column, washing with appropriate buffers, and final elution of purified DNA. DNA concentrations were measured using a Qubit™ 4 Fluorometer with the Qubit™ 1X dsDNA BR Assay Kit (Invitrogen, Thermo Fisher Scientific India Pvt. Ltd., Mumbai, India; Cat. No. Q33265), following the manufacturer's instructions. Briefly, 2 µL of each DNA sample was mixed with 198 µL of Qubit™ 1X dsDNA BR Working Solution in a Qubit™ assay tube. The mixture was vortexed for 3–5 s and incubated at room temperature for 2 min before fluorescence measurement using the 1X dsDNA BR assay mode. The DNA samples were subsequently diluted 1:10 with molecular biology-grade water and used as templates for endpoint PCR and qPCR.

2.3 Endpoint Polymerase Chain Reaction

Endpoint PCR was performed before qPCR to optimize annealing temperatures. A total of five bacterial primer sets targeting the 16S ribosomal RNA (rRNA) gene (Table 1) and three fungal primer sets targeting the internal transcribed spacer (ITS) region (Table 2) were selected to detect selected bacterial and fungal targets. Each 50 µL endpoint PCR reaction contained 5.0 µL of 10× iTaq buffer, 1.5 µL of 50 mM $MgCl_2$, 1.0 µL of 10 mM dNTP mix, 0.25 µL of iTaq DNA polymerase (5 U/µL), forward and reverse primers at final concentrations of 0.3 µM each, and 5 ng of DNA template. Molecular biology-grade water was added to a final reaction volume of 50 µL. Thermal cycling comprised an initial denaturation at 95 °C for 5 min, followed by 40 cycles of denaturation at 95 °C for 15 s, annealing at a gradient from 47 °C to 56 °C for 15 s, and extension at 72 °C for 20 s, with a final extension at 72 °C for 5 min. Amplified products were resolved on 2% agarose gels at 90 V for 60 min and visualized using a gel documentation system to confirm amplicon size and determine the optimal annealing temperature for each primer pair.

Table 1. Bacterial primers used in this study

Taxa	Primers	Primer Sequence (5′–3′)	Amplicon Size (bp)	Reference
All bacteria	F- Bakt_341F R- Bakt_805R	CCTACGGGNGGCWGCAG GACTACHVGGGTATCTAATCC	464	Herlemann et al., 2011
Alphaproteobacteria	F- Eub338 R- Alf685	ACTCCTACGGGAGGCAGCAG TCTACGRATTTCACCYCTAC	365	Fierer et al., 2005; Lane, 1991
Betaproteobacteria	F- Eub338 R- Bet680	ACTCCTACGGGAGGCAGCAG TCACTGCTACACGYG	360	Fierer et al., 2005; Lane, 1991; Overmann et al., 1999
Acidobacteria	F- Acid31 R- Eub518	GATCCTGGCTCAGAATC ATTACCGCGGCTGCTGG	180	Barns et al., 1999; Fierer et al., 2005; Muyzer et al., 1993
Actinobacteria	F- Actino235 R- Eub518	CGCGGCCTATCAGCTTGTTG ATTACCGCGGCTGCTGG	300	Fierer et al., 2005; Muyzer et al., 1993; Stach et al., 2003

Note: F represents the forward primer, and R represents the reverse primer. In primer design, N, H, K, R, S, V, W, and Y are IUPAC codes representing ambiguous bases. N represents any of the four bases, Y represents a pyrimidine (C or T), R represents a purine (A or G), K represents a keto base (G or T), S represents a strong base (G or C), W represents a weak base (A or T), H represents A, C, or T, V represents A, C, or G.

Table 2. Fungal primers used in this study

Taxa	Primers	Primer Sequence (5′–3′)	Amplicon Size (bp)	Reference
All fungi	F- ITS1F R- ITS4	CTTGGTCATTTAGAGGAAGTAA TCCTCCGCTTATTGATATGC	574–789	Op De Beeck et al., 2014; White et al., 1990
Ascomycota	F- ITS1F R- ITS4A	CTTGGTCATTTAGAGGAAGTAA CGCCGTTACTGGGGCAATCCTG	650	Larena et al., 1999
Basidiomycota	F- ITS1F R- ITS4B	CTTGGTCATTTAGAGGAAGTAA CAGGAGACTTGTACACGGTCCAG	500	Gardes & Bruns, 1993

Note: F represents the forward primer, and R represents the reverse primer.

2.4 Quantitative Polymerase Chain Reaction

The qPCR was performed using all eight primer sets to compare qPCR-derived target-gene signals between the sampled forest and agrochemically treated agricultural soils. The reaction mixture was 20 µl, consisting of 10 µL of 2× iTaq™ Universal SYBR® Green Supermix, 0.6 µL each of 10 µM forward and reverse primers (final concentration of 0.3 µM each), and approximately 5 ng of template DNA, with molecular biology-grade water added to make up the final volume. For each primer set, five reaction tubes were assembled: two replicates for the agrochemically treated soil, two replicates for the forest soil, and one no-template control. Thermal cycling began with a 5-min denaturation step at 95 °C, followed by 40 amplification cycles. Each cycle consisted of 15 s at 95 °C, 15 s at the primer-specific optimized annealing temperature, and 20 s at 72 °C for extension. A melting curve was generated from 65 °C to 95 °C in 0.5 °C increments (5 s per step) to assess amplification specificity.

2.5 Statistical Analysis

The qPCR data were analyzed by comparing cycle threshold (Ct) values for each target between agricultural and forest soil samples. For each target, ΔCt was calculated as the difference between the mean Ct values of the

two sampled sites: $\Delta Ct = \overline{Ct}_{agricultural} - \overline{Ct}_{forest}$. The estimated fold difference in qPCR-derived target-gene signals was calculated as $2^{-\Delta Ct}$. This calculation assumes approximately 100% amplification efficiency, which was not experimentally verified in the present study. Values greater than 1 indicated higher target-gene signals, whereas values less than 1 indicated lower target-gene signals in agricultural soil relative to forest soil. A similar calculation approach has been used in soil microbial qPCR research (Pei et al., 2026). Ct values from the two composite soil samples per site were used for exploratory statistical comparisons. Differences between agricultural and forest soil samples were evaluated using Welch's two-sample *t*-test, with statistical significance defined as $p \leq 0.05$. Because each land-use type was represented by only one sampling site, these comparisons were restricted to the two sampled sites and do not establish generalizable land-use effects.

3. Results

3.1 Optimization of Annealing Temperature

Annealing temperatures for all primer sets were empirically determined by conducting independent PCR runs across a temperature gradient ranging from 47 °C to 56 °C (1 °C increment). The optimal annealing temperature for each primer pair was defined as the lowest temperature yielding a single, specific amplicon of the expected size with minimal non-specific amplification and primer-dimer formation (Table 3). The primer sets targeting *Ascomycota*, *Basidiomycota*, and total fungi showed the best amplification at 50 °C, with reduced non-specific products and primer-dimer bands at this temperature. Similarly, the primer pairs for all bacteria and *Alphaproteobacteria* demonstrated optimal specificity at 50 °C, while the primer pair targeting *Actinobacteria* produced discrete bands of the anticipated size (~300 bp) at 53 °C across both agrochemically treated and forest soil samples. *Betaproteobacteria* and *Acidobacteria* primers, which possess comparatively lower melting temperatures, required a reduced annealing temperature of 47 °C to obtain specific amplification. *Basidiomycota*-specific amplification was not detected in the agricultural soil samples at any of the annealing temperatures tested during endpoint PCR, a finding consistent with the subsequent qPCR results. Although residual non-specific amplification and primer-dimer formation could not be eliminated, these were reduced following optimization. Collectively, these optimizations successfully mitigated non-specific amplification and primer-dimer formation, thereby supporting the subsequent qPCR analysis.

Table 3. Primer characteristics and optimal annealing temperatures determined by gradient PCR

Taxa	Forward Primer	Tm (°C)	Reverse Primer	Tm (°C)	Amplicon Size (bp)	Optimal Ta (°C)
All bacteria	Bakt_341F	56.3–60.2	Bakt_805R	48.8–53.0	464	50
Alphaproteobacteria	Eub338	60.4	Alf685	48.5–52.7	365	50
Betaproteobacteria	Eub338	60.4	Bet680	47.0–50.8	360	47
Acidobacteria	Acid31	48.5	Eub518	57.8	180	47
Actinobacteria	Actino235	59.1	Eub518	57.8	300	53
All fungi	ITS1F	49.9	ITS4	52.1	574–789	50
Ascomycota	ITS1F	49.9	ITS4A	62.7	650	50
Basidiomycota	ITS1F	49.9	ITS4B	59.0	500	50

Note: PCR = polymerase chain reaction. Primer identifiers correspond to those listed in Tables 1 and 2; Tm = primer melting temperature; Ta = optimal annealing temperature.

3.2 Differences in Bacterial and Fungal Target-Gene Signals Between Agricultural and Forest Soils

The qPCR analysis was employed to compare the qPCR-derived signals of selected bacterial and fungal taxa between soil samples from the agrochemically treated agricultural site and the nearby forest site. Relative differences in target-gene signals were estimated using the $2^{-\Delta Ct}$ method, as described in Section 2.5. Values greater than 1 indicate higher target-gene signals, and values less than 1 indicate lower target-gene signals, in the sampled agricultural soil relative to the forest soil. Statistical significance of inter-group differences in mean Ct values was assessed using Welch's *t*-test, with significance defined at $p \leq 0.05$.

The total bacterial qPCR-derived signal in sampled agricultural soil was slightly lower (0.84-fold) than that in forest soil; this difference was not statistically significant ($p = 0.531$). Among the selected bacterial taxa, *Alphaproteobacteria* showed a lower qPCR signal in the agricultural soil (0.57-fold; $p = 0.002$), as did *Acidobacteria*, which exhibited the most pronounced decline among the bacterial taxa (0.26-fold; $p = 0.008$). In contrast, *Betaproteobacteria* showed a higher qPCR signal in the agrochemically treated soil (1.97-fold; $p = 0.001$). Although the signal for *Actinobacteria* was also slightly higher (1.09-fold) in sampled agricultural soil, this difference was not statistically significant between the two sampled sites ($p = 0.768$). The corresponding results for the selected bacterial taxa are summarized in Table 4.

Table 4. ΔCt values and corresponding fold change for bacterial taxa between sampled agricultural and forest soils

Taxa	Mean Ct (Agricultural Soil)	Mean Ct (Forest Soil)	ΔCt	Fold Change ($2^{-\Delta Ct}$)	*p*-Value	Remarks
All bacteria	26.86	26.60	0.26	0.84	0.531	Lower in agricultural
Alphaproteobacteria	27.55	26.74	0.81	0.57	0.002	Lower in agricultural *
Betaproteobacteria	29.30	30.28	-0.98	1.97	0.001	Higher in agricultural *
Acidobacteria	29.03	27.08	1.95	0.26	0.008	Lower in agricultural *
Actinobacteria	25.73	25.86	-0.13	1.09	0.768	Higher in agricultural

Note: Two composite soil samples were analyzed per site ($n = 2$ per site), with each composite consisting of five subsamples. ΔCt = difference in mean Ct values between agricultural and forest soil samples (agricultural − forest); positive values indicate lower target-gene signals in agricultural soil, whereas negative values indicate higher signals. * $p \leq 0.05$ (unadjusted Welch's *t*-test).

The selected fungal targets exhibited larger qPCR-derived differences between the two sampled sites than most of the bacterial targets. The total fungal qPCR signal was substantially lower in sampled agricultural soil (0.06-fold; $p = 0.007$), representing the largest magnitude of change among the targets with quantifiable Ct values. The signal for *Ascomycota* was also lower in the agricultural soil (0.40-fold; $p = 0.050$). *Basidiomycota* displayed a distinct detection pattern: target-specific amplification was detected in sampled forest soil (mean Ct = 31.20) but was not detected under the applied qPCR conditions in the agricultural soil replicates, precluding quantitative fold-change comparison. The corresponding results for the selected fungal taxa are summarized in Table 5.

Table 5. ΔCt values and corresponding fold change for fungal taxa between sampled agricultural and forest soils

Taxa	Mean Ct (Agricultural Soil)	Mean Ct (Forest Soil)	ΔCt	Fold Change ($2^{-\Delta Ct}$)	*p*-Value	Remarks
All fungi	33.40	29.22	4.18	0.06	0.007	Lower in agricultural *
Ascomycota	22.81	21.48	1.33	0.40	0.0503	Lower in agricultural
Basidiomycota	ND	31.20	—	—	—	Not detected in agricultural soil

Note: Two composite soil samples were analyzed per site ($n = 2$ per site), with each composite consisting of five subsamples. ND = not detected under the applied assay conditions; "—" = not calculated because the target was not detected in agricultural soil; ΔCt = difference in mean Ct values between agricultural and forest soil samples (agricultural − forest); positive values indicate lower target-gene signals in agricultural soil, whereas negative values indicate higher signals. * $p \leq 0.05$ (unadjusted Welch's *t*-test).

The overall differences in qPCR-derived target-gene signals between the sampled agricultural and forest soils are illustrated in Figure 1 using $\log_2$ fold changes and in Figure 2 using mean Ct values across the selected bacterial and fungal taxa.

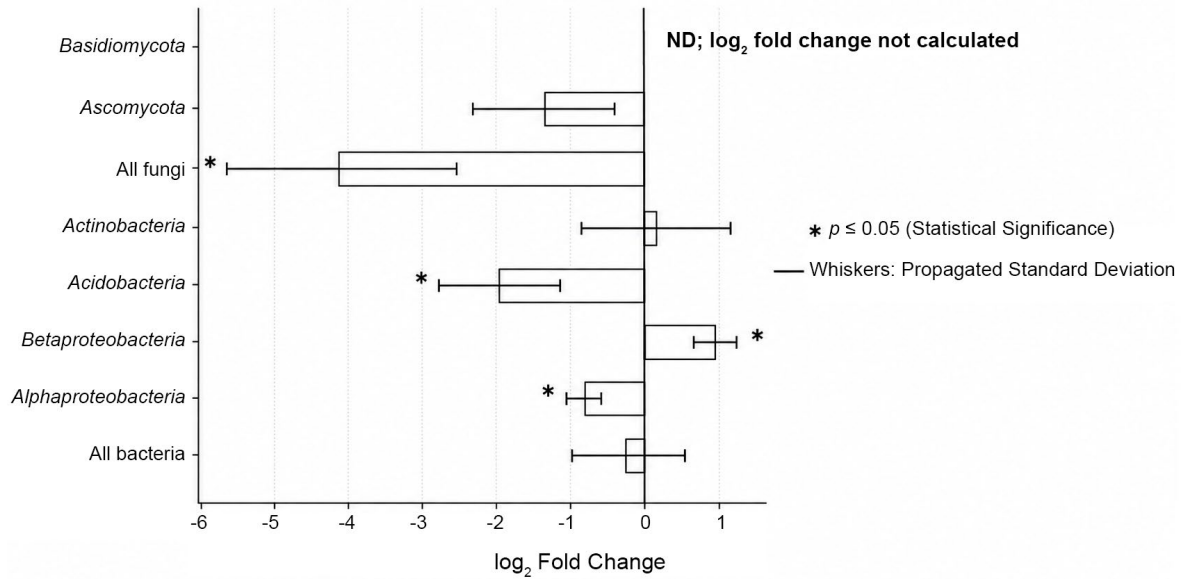


Figure 1. Horizontal bar chart depicting $\log_2$ fold changes in qPCR-derived target-gene signals between sampled agricultural and forest soils

Note: qPCR = quantitative polymerase chain reaction; ND = not detected under the applied assay conditions. Positive values indicate higher target-gene signals in agricultural soil, whereas negative values indicate higher signals in forest soil. Error bars represent propagated standard deviation. Asterisks (*) indicate statistically significant differences ($p \leq 0.05$, unadjusted Welch's *t*-test).

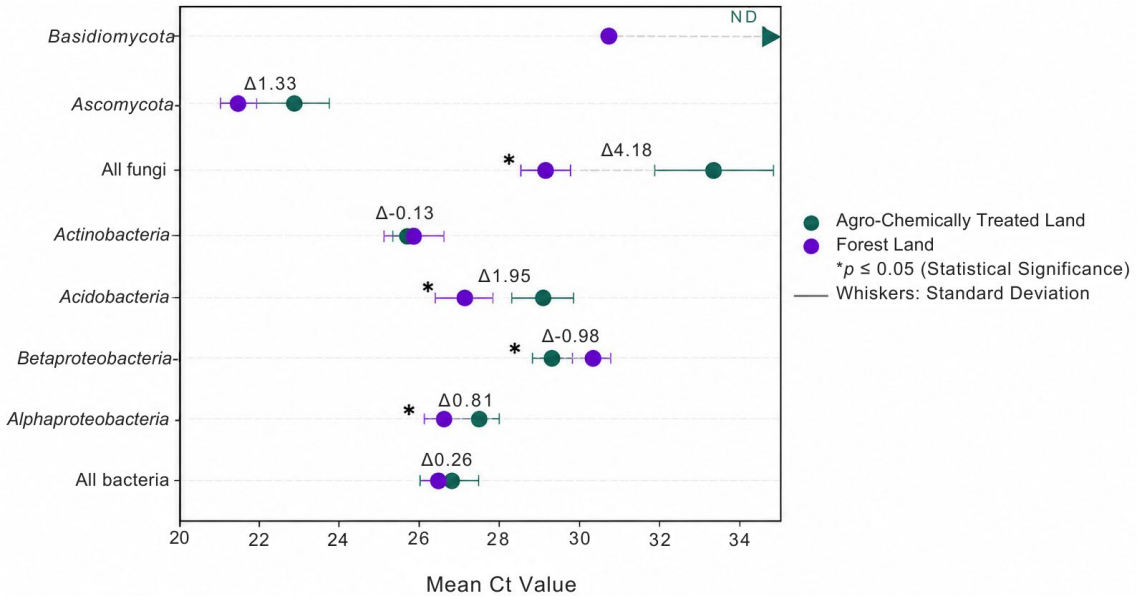


Figure 2. Dumbbell plot comparing mean Ct values of selected bacterial and fungal taxa between sampled agricultural and forest soils

Note: Ct = cycle threshold; ND = not detected under the applied assay conditions. Horizontal dashed lines connect the two site means for each taxon. Lower Ct values indicate higher target-gene signals within the same assay under comparable reaction conditions. Asterisks (*) indicate statistically significant differences ($p \leq 0.05$, unadjusted Welch's *t*-test).

4. Discussion

In this study, qPCR-derived target-gene signals for selected bacterial and fungal taxa were compared between an agrochemically managed agricultural field and a forest site at a similar elevation. The qPCR-derived results indicated lower target-gene signals for total bacteria and total fungi in the agricultural soil samples relative to the forest samples. However, the difference in the total bacterial signal was not statistically significant. Soil microbial communities vary across different land-use types, possibly due to anthropogenic activities and differences in land management (Sun et al., 2024). In particular, reduced soil tillage intensity has been associated with greater fungal biomass (Sae-Tun et al., 2022). The agricultural site investigated in this study was a farmer-managed cauliflower farm where insecticides, fungicides, herbicides, chemical fertilizers, and organic manure were commonly applied, whereas the forest site was undisturbed and had not been subjected to cultivation or agrochemical inputs. These contrasting management practices may have contributed to the differences in qPCR-derived target-gene signals observed between the two sites. However, because the present study did not quantify pesticide residues, soil physicochemical properties, and other environmental variables known to influence microbial communities, it is not possible to attribute the observed differences directly to agrochemical use. Therefore, although the findings are consistent with previous reports that land-use and management practices influence soil microbial communities, further studies incorporating multiple independent sampling sites, soil physicochemical data, and measurements of agrochemical residues are required to identify the specific factors responsible for the observed differences.

The qPCR-derived signal for *Acidobacteria* was lower (0.26-fold), while that for *Actinobacteria* was slightly higher (1.09-fold) at the sampled agricultural site compared to the forest site, although the latter difference was not statistically significant. This is consistent with findings from previous studies. Xu et al. (2025) found that *Acidobacteria* and *Proteobacteria* had higher relative abundances in forest lands, while *Actinobacteria* had higher abundance in croplands. *Acidobacteria* are known to be oligotrophic bacteria and can respond to changes in nutrient availability and soil pH (Lladó et al., 2017). The abundance of *Actinobacteria* is associated with land-use practices and soil moisture conditions (Sun et al., 2024). A meta-analysis of tropical land-use conversion studies reported decreases in the relative abundances of *Acidobacteria* and *Proteobacteria* and an increase in *Actinobacteria* in agricultural soils relative to forest soils (Petersen et al., 2019). Although our results showed similar patterns, the observed differences cannot be causally attributed to land-use type or agrochemical use, as this study compared only one agricultural site with one forest site, and soil physicochemical properties, vegetation, management history, and agrochemical exposure were not systematically characterized or controlled.

The qPCR-derived signal for *Alphaproteobacteria* was higher in the sampled forest soil, whereas that for *Betaproteobacteria* was higher in the sampled agricultural soil. Previous research has shown that fertilization practices can be associated with shifts in bacterial community composition (Francioli et al., 2016). However, because soil nutrient availability and microbial functional traits were not assessed in this study, these ecological interpretations should be regarded as literature-based hypotheses rather than mechanisms demonstrated by the

present data.

Lower qPCR-derived signals for total fungi and *Ascomycota* were observed in sampled agricultural soil. Notably, *Basidiomycota*-specific amplification was not detected in the agricultural samples under the applied qPCR conditions. Substantial reductions in *Basidiomycota* following the conversion of forest land to agricultural land and under intensive agricultural management have been reported (Navarro-Noya et al., 2021). Information obtained from the farmer in this study confirmed that fungicides formed part of the routine crop management practices in the sampled agricultural field. However, fungal community composition is known to be influenced by multiple factors, including vegetation type, land-use practices, soil physicochemical properties, and agrochemical use (Boddy et al., 2007; Wightwick et al., 2010). Therefore, although fungicide application may have contributed to the observed non-detection of *Basidiomycota* in the agricultural samples, this relationship cannot be confirmed by the present study because pesticide residues, soil physicochemical properties, and other management variables were not evaluated.

This research provides preliminary evidence of differences in qPCR-derived target-gene signals between the sampled agrochemically managed agricultural field and nearby forest site in the Panchkhal region of Nepal. However, the findings should be interpreted with some limitations. The study was conducted at only one site for each land-use type, and subsamples were pooled into two composite samples per site, which may have obscured within-site spatial heterogeneity. The absence of independent site-level replication also limits inference about land-use types. Furthermore, pesticide residues, microbial physiological activity, and soil physicochemical properties (including pH, moisture content, soil organic carbon, total nitrogen, available phosphorus, electrical conductivity, and soil texture) were not measured in the present study. Consequently, the mechanisms underlying the observed differences could not be directly determined. The use of taxon-specific qPCR limited the analysis to selected microbial groups and did not capture the full diversity or functional potential of the soil microbiome, and the possibility of residual PCR inhibition cannot be completely excluded. Because standard-curve slopes, amplification efficiencies, coefficients of determination (R^2), analytical sensitivity, inhibition assessments, and extraction controls were not assessed during the present study, our results should be interpreted as exploratory comparisons of qPCR-derived target-gene signals between the two sampled sites rather than validated estimates of absolute gene-copy numbers or soil microbial abundance.

5. Conclusions

This exploratory study identified differences in qPCR-derived target-gene signals for selected bacterial and fungal taxa between one agrochemically managed agricultural site and one nearby forest site in Panchkhal, Nepal. Within the sampled sites, the forest soil showed higher qPCR-derived signals for total fungi, *Alphaproteobacteria*, *Acidobacteria*, and *Ascomycota*, while *Basidiomycota*-specific amplification was detected only in the forest samples. The agricultural soil showed higher qPCR-derived signals for *Betaproteobacteria* and *Actinobacteria*, although the latter difference was not statistically significant. Because the study was based on a single agricultural site and a single forest site, these findings should be interpreted as site-specific observations rather than representative of agricultural and forest soils throughout Panchkhal or Nepal. Furthermore, the observational design and the absence of measurements of pesticide residues, soil physicochemical properties, and other environmental variables preclude conclusions regarding the specific effects of agrochemical management on the observed qPCR-derived signals. Future studies involving multiple independent sampling sites, comprehensive environmental measurements, and high-throughput sequencing approaches are needed to distinguish the effects of land-use, agrochemical management, and other environmental factors on soil microbial communities.

Author Contributions

Conceptualization, A.A., Y.B., D.G., S.K., D.P., and A.D.; methodology, A.A., Y.B., D.G., S.K., D.P., and A.D.; software, A.A., Y.B., D.G., S.K., and D.P.; validation, A.D.; formal analysis, A.A., Y.B., D.G., S.K., D.P., and A.D.; investigation, A.A., Y.B., D.G., S.K., and D.P.; resources, A.D.; data curation, A.A., Y.B., D.G., S.K., D.P., and A.D.; writing—original draft preparation, A.A., Y.B., D.G., S.K., and D.P.; writing—review and editing, A.D.; visualization, A.D.; supervision, A.D.; project administration, A.D. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgments

The authors would like to thank the Department of Biotechnology at Kathmandu University for providing

laboratory space and research materials.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative AI tools were used solely for language editing and improvement of readability. The authors reviewed and edited the output as necessary and take full responsibility for the content of the manuscript.

References

Barns, S. M., Takala, S. L., & Kuske, C. R. (1999). Wide distribution and diversity of members of the bacterial kingdom Acidobacterium in the environment. *Appl. Environ. Microbiol.*, *65*(4), 1731–1737. https://doi.org/10.1128/AEM.65.4.1731-1737.1999.

Boddy, L., Frankland, J., & van West, P. (2007). *Ecology of Saprotrophic Basidiomycetes*. Academic Press.

Fierer, N., Jackson, J. A., Vilgalys, R., & Jackson, R. B. (2005). Assessment of soil microbial community structure by use of taxon-specific quantitative PCR assays. *Appl. Environ. Microbiol.*, *71*(7), 4117–4120. https://doi.org/10.1128/AEM.71.7.4117-4120.2005.

Francioli, D., Schulz, E., Lentendu, G., Wubet, T., Buscot, F., & Reitz, T. (2016). Mineral vs. organic amendments: Microbial community structure, activity and abundance of agriculturally relevant microbes are driven by long-term fertilization strategies. *Front. Microbiol.*, *7*, 1446. https://doi.org/10.3389/fmicb.2016.01446.

Gardes, M. & Bruns, T. D. (1993). ITS primers with enhanced specificity for basidiomycetes—Application to the identification of mycorrhizae and rusts. *Mol. Ecol.*, *2*(2), 113–118. https://doi.org/10.1111/j.1365-294X.1993.tb00005.x.

GC, S. & Neupane, J. (2019). Pesticides use in Nepal and its effects on human health—A review. *Acta Sci. Agric.*, *3*(11), 114–117. https://doi.org/10.31080/ASAG.2019.03.0694.

Ghimire, K. & GC, A. (2018). Trend of pesticide use in Nepal. *J. Plant Prot. Soc.*, *5*, 32–42. https://doi.org/10.3126/jpps.v5i0.37755.

Herlemann, D. P. R., Labrenz, M., Jürgens, K., Bertilsson, S., Waniek, J. J., & Andersson, A. F. (2011). Transitions in bacterial communities along the 2000 km salinity gradient of the Baltic Sea. *ISME J.*, *5*(10), 1571–1579. https://doi.org/10.1038/ismej.2011.41.

Lane, D. J. (1991). 16S/23S rRNA sequencing. In E. Stackebrandt & M. Goodfellow (Eds.), *Nucleic Acid Techniques in Bacterial Systematics* (pp. 115–175). John Wiley & Sons.

Larena, I., Salazar, O., González, V., Julián, M. C., & Rubio, V. (1999). Design of a primer for ribosomal DNA internal transcribed spacer with enhanced specificity for ascomycetes. *J. Biotechnol.*, 75(2–3), 187–194. https://doi.org/10.1016/S0168-1656(99)00154-6.

Lladó, S., López-Mondéjar, R., & Baldrian, P. (2017). Forest soil bacteria: Diversity, involvement in ecosystem processes, and response to global change. Microbiol. *Mol. Biol. Rev.*, *81*(2), e00063-16. https://doi.org/10.1128/MMBR.00063-16.

Muyzer, G., de Waal, E. C., & Uitterlinden, A. G. (1993). Profiling of complex microbial populations by denaturing gradient gel electrophoresis analysis of polymerase chain reaction-amplified genes coding for 16S rRNA. *Appl. Environ. Microbiol.*, *59*(3), 695–700. https://doi.org/10.1128/AEM.59.3.695-700.1993.

Navarro-Noya, Y. E., Montoya-Ciriaco, N., Muñoz-Arenas, L. C., Hereira-Pacheco, S., Estrada-Torres, A., & Dendooven, L. (2021). Conversion of a high-altitude temperate forest for agriculture reduced alpha and beta diversity of the soil fungal communities as revealed by a metabarcoding analysis. *Front. Microbiol.*, *12*, 667566. https://doi.org/10.3389/fmicb.2021.667566.

Némethová, E., Řezáč, M., Gryndler, M., Abeyawardana, O. A. J., & Řezáčová, V. (2026). Soil pH mediates the impact of pesticides on bacterial communities, diversity, and abundance. *Front. Microbiol.*, *16*, 1670425. https://doi.org/10.3389/fmicb.2025.1670425.

Op De Beeck, M., Lievens, B., Busschaert, P., Declerck, S., Vangronsveld, J., & Colpaert, J. V. (2014). Comparison and validation of some ITS primer pairs useful for fungal metabarcoding studies. *PLoS ONE, 9*(6), e97629. https://doi.org/10.1371/journal.pone.0097629.

Osinuga, O. A., Aduloju, A. B., & Oyegoke, C. O. (2023). Impact of agrochemicals application on soil quality indicators and trace elements level of wetlands under different uses. *J. Trace Elem. Miner.*, *5*, 100090. https://doi.org/10.1016/j.jtemin.2023.100090.

Overmann, J., Coolen, M. J. L., & Tuschak, C. (1999). Specific detection of different phylogenetic groups of chemocline bacteria based on PCR and denaturing gradient gel electrophoresis of 16S rRNA gene fragments.

Arch. Microbiol., *172*(2), 83–94. https://doi.org/10.1007/s002030050744.

Paudel, L., Neupane, R., Shrestha, L., Budhathoki, L., & Paudel, K. R. (2024). Knowledge on chemical pesticide use among farmers exposed to pesticides in Panchkhal Municipality, Kavrepalanchok, Nepal. *Adv. Public Health*, *2024*(1), 3110674. https://doi.org/10.1155/2024/3110674.

Pei, X., Zhang, N., Deng, X., Li, R., Wang, Y., Wang, Y., Huang, W., Yue, Y., Geisen, S., Gao, Z., et al. (2026). Biofertilizer induces soil disease suppression by activating pathogen suppressive protist taxa. *NPJ Biofilms Microbiomes*, *12*, 31. https://doi.org/10.1038/s41522-025-00897-2.

Petersen, I. A. B., Meyer, K. M., & Bohannan, B. J. M. (2019). Meta-analysis reveals consistent bacterial responses to land use change across the tropics. *Front. Ecol. Evol.*, *7*, 391. https://doi.org/10.3389/fevo.2019.00391.

Sae-Tun, O., Bodner, G., Rosinger, C., Zechmeister-Boltenstern, S., Mentler, A., & Keiblinger, K. (2022). Fungal biomass and microbial necromass facilitate soil carbon sequestration and aggregate stability under different soil tillage intensities. *Appl. Soil Ecol.*, *179*, 104599. https://doi.org/10.1016/j.apsoil.2022.104599.

Sapkota, K. R., Sapkota, S., Sapkota, S., & Katuwal, K. (2020). Pesticides handling practices among potato growers in Kavrepalanchok, Nepal. *J. Agric. Nat. Res.*, *3*(1), 77–87. https://doi.org/10.3126/janr.v3i1.27093.

Srivastav, A. L. (2020). Chemical fertilizers and pesticides: Role in groundwater contamination. In M. N. V. Prasad (Ed.), *Agrochemicals Detection, Treatment and Remediation: Pesticides and Chemical Fertilizers* (pp. 143–159). Butterworth-Heinemann. https://doi.org/10.1016/B978-0-08-103017-2.00006-4.

Stach, J. E. M., Maldonado, L. A., Ward, A. C., Goodfellow, M., & Bull, A. T. (2003). New primers for the class Actinobacteria: Application to marine and terrestrial environments. *Environ. Microbiol.*, *5*(10), 828–841. https://doi.org/10.1046/j.1462-2920.2003.00483.x.

Sun, Z., Sun, C., Zhang, T., Liu, J., Wang, X., Feng, J., Li, S., Tang, S., & Jin, K. (2024). Soil microbial community variation among different land use types in the agro-pastoral ecotone of northern China is likely to be caused by anthropogenic activities. *Front. Microbiol.*, *15*, 1390286. https://doi.org/10.3389/fmicb.2024.1390286.

White, T. J., Bruns, T., Lee, S., & Taylor, J. (1990). Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. In M. A. Innis, D. H. Gelfand, J. J. Sninsky, & T. J. White (Eds.), *PCR Protocols: A Guide to Methods and Applications* (pp. 315–322). Academic Press. https://doi.org/10.1016/B978-0-12-372180-8.50042-1.

Wightwick, A., Walters, R., Allinson, G., Reichman, S., & Menzies, N. (2010). Environmental risks of fungicides used in horticultural production systems. In O. Carisse (Ed.), *Fungicides* (pp. 273–304). InTech. https://doi.org/10.5772/13032.

Xu, L., Cheng, D., Feng, L., Lu, X., Ruffell, S., & Wang, H. (2025). Bacterial community composition is an important predictor of surface soil fertility across different land use types: A case study in the Three Gorges Reservoir area. *PeerJ*, *13*, e18959. https://doi.org/10.7717/peerj.18959.